

CCE-EDUSAT SESSION FOR COMPUTER FUNDAMENTALS

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Topic : **Input/Output Devices**

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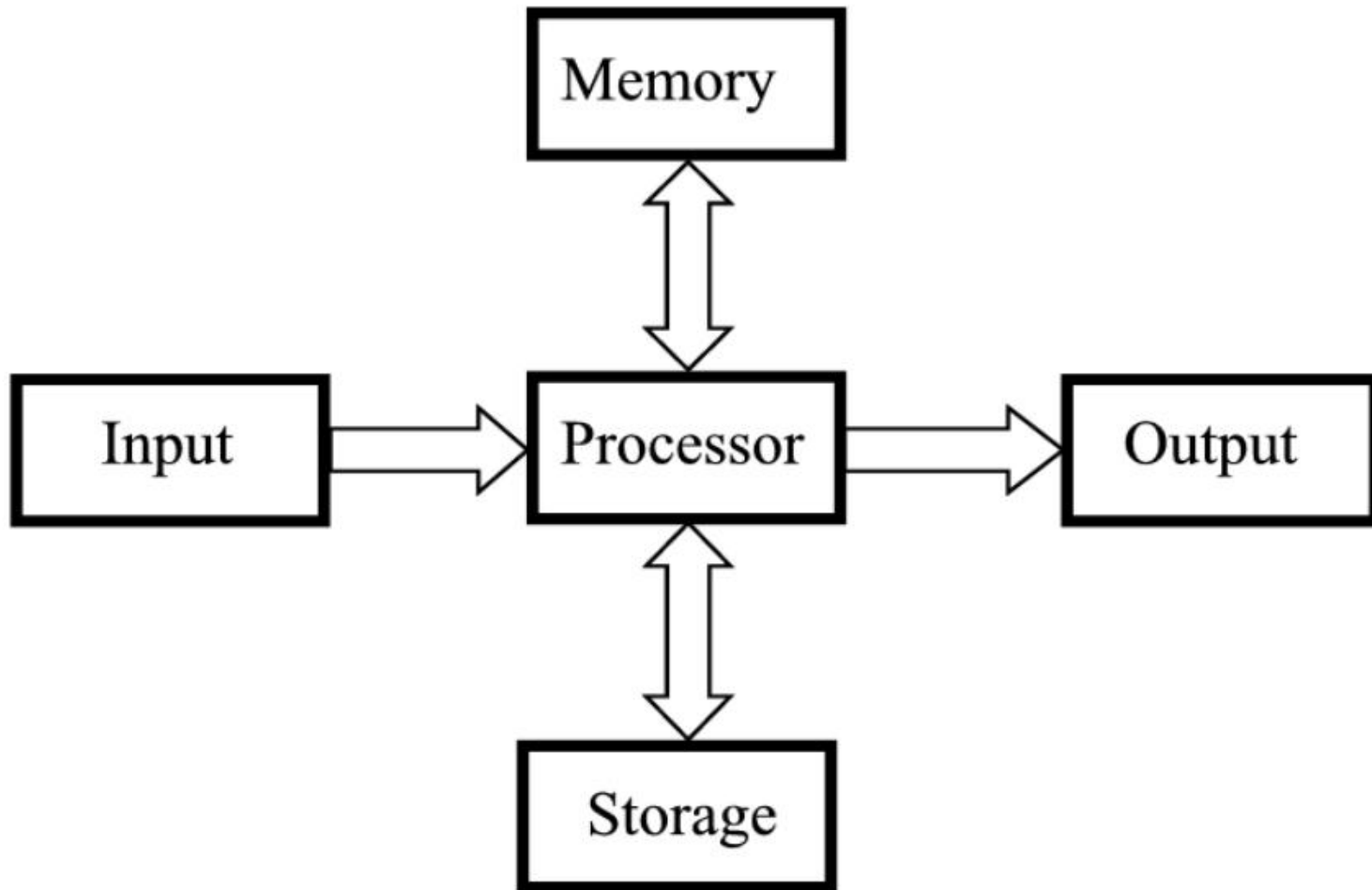
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- Input Devices
 - Keyboard, mouse, joystick, scanners, digital camera, bar code reader, touch Sreeen, Speech input device (microphone)
- Output Devices
 - Monitor , Speaker, Printers (different types)

1.Introduction



What's available for input...

- touch - fingers, feet, breath
- sound - voice, other sounds
- gesture
- gaze
- brainwaves...

and output...

- textual information
- visual images - photos, diagrams, icons
- moving images
- sounds - music, voice
- Etc..

2. Input Devices

- Used by a person to communicate to a computer.



Person to
computer



Output Devices

- Displays information from the computer to a person.



Input Devices

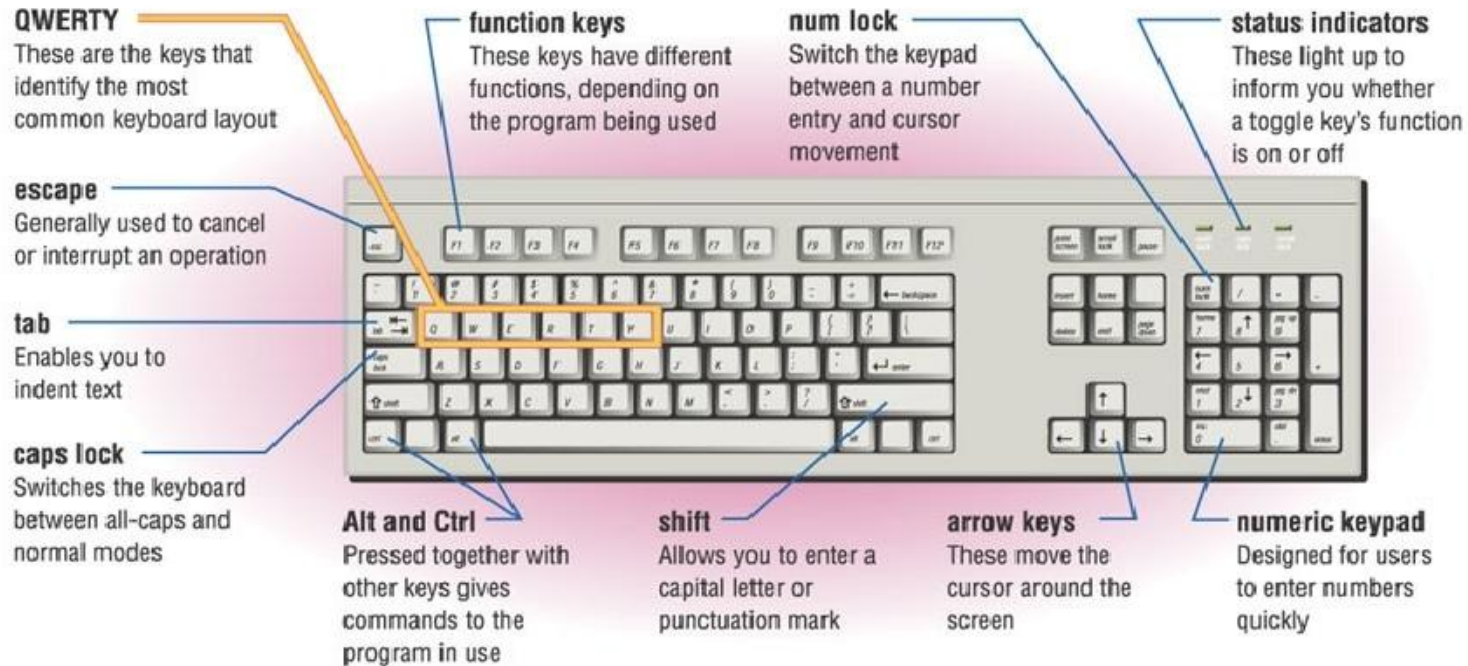
- Keyboard.
- Mouse
- Microphone
- Digital Camera
- Scanner

The Keyboard

The most commonly used input device is the keyboard on which data is entered by manually keying in or typing certain keys. A keyboard typically has 101 or 105 keys.



Keyboard



- The keyboard allows the computer user to enter words, numbers, punctuation, symbols, and special function commands into the computer's memory.

The Mouse

Is a pointing device which is used to control the movement of a mouse pointer on the screen to make selections from the screen. A mouse has one to five buttons. The bottom of the mouse is flat and contains a mechanism that detects movement of the mouse.



Pointing devices - direct

- **Touch screens**



Touch screens

- Often used for applications with occasional use, for example
- Bank ATMs, Information Kiosks, etc.
- No extra hardware - used for input and for output
- Can be precise to 1 pixel
- Good for menu choice - not so good for other functions
- Intuitive to use

Touch screens

- BUT
- Tiring if at wrong angle (needs to be 30-45% from horizontal)
- Get greasy, jammy
- Finger can obscure screen
- Alternative - use stylus to touch screen, or lightpen

Indirect Pointing Devices

- Need more cognitive processing than direct methods, but can be more efficient
- mouse
- tracker ball
- track point
- touchpad...

Indirect pointing devices - other

- Tracker ball, trackpad, trackpoint
- Less space on desktop
- Good in moving environments, e.g. car, train



Indirect pointing devices - other

- Joystick
- The main use of a joystick is to play computer games by controlling the way that something moves on the screen.



Microphones - Speech Recognition

- Use a microphone to talk to your computer
 - Add a sound card to your computer
 - Sound card digitizes audio input into 0/1s
- A speech recognition program can process the input and convert it into machine-recognized commands or input

Audio Input: Speech Recognition

- **Speech recognition** is a type of input in which the computer recognizes words spoken into a microphone.
- Special software and a microphone are required.
- Latest technology uses continuous speech recognition where the user does not have to pause between words.



Microphones - Speech Recognition



1 "Up" spoken word



2 Waveform of word

3 Microphone



4 Speech recognition board

01001100

5 Waveform converted into binary code

6 Binary code compared to stored vocabulary

01001001 = us

01001100 = up

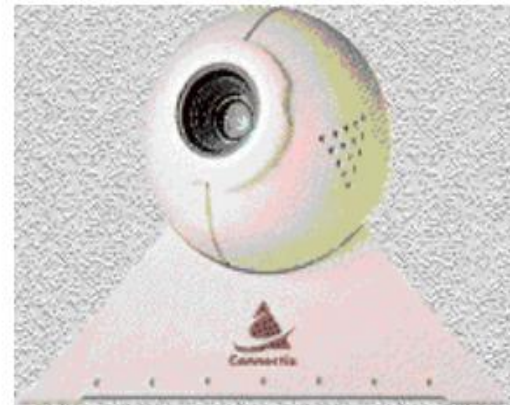
00011010

7 Word displayed on screen



Digital camera

A **digital camera** can store many more pictures than an ordinary camera. Pictures taken using a digital camera are stored inside its memory and can be transferred to a computer by connecting the camera to it. A digital camera takes pictures by converting the light passing through the lens at the front into a digital image.



Scanner

A scanner can be used to input pictures and text into a computer. There are two main types of scanner; Hand-held and Flat-bed.



Light pen

- A light pen is a small 'pen-shaped' wand, which contains light sensors.
- It is used to choose objects or commands on the screen either by pressing it against the surface of the screen or by pressing a small switch on its side.
- A signal is sent to the computer, which then works out the light pen's exact location on the screen.
- The advantage of a light pen is that it doesn't need a special screen or screen coating.

Bar codes



- A bar code is a set of lines of different thicknesses that represent a number
- Bar Code Readers are used to input data from bar codes. Most products in shops have bar codes on them
- Bar code readers work by shining a beam of light on the lines that make up the bar code and detecting the amount of light that is reflected back

2. Output Types

- Text output
- Graphics output
- Video output
- Audio output

Output Devices: Engaging our Senses



- Output devices are peripheral devices that enable us to view or hear the computer's processed data.
 - **Visual output** – Text, graphics, and video
 - **Audio output** – Sounds, music, and synthesized speech

Output Devices

- Monitors
- Printers
- Speakers
- Data Projectors

Monitors



CRT



LCD

- A monitor is a peripheral device which displays computer output on a screen.
- Screen output is referred to as **soft copy**.
- Types of monitors:
 - **Cathode-ray tube (CRT)**
 - **Liquid Crystal Display (LCD or flat-panel)**

Monitors

- CRT

- cathode ray tube

- electron gun shoots a stream of electrons at a special phosphor-coated screen
 - on impact, the phosphor flares up for a fraction of a second
 - electron gun sweeps across the screen many times a second



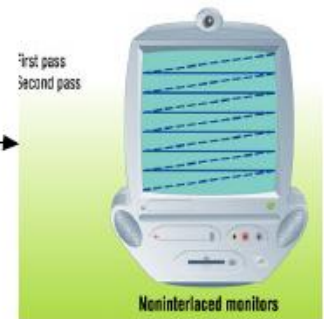
- LCD

- liquid crystal display

- one of several types of “flat-panel” displays
 - forms output by solidifying crystals and “backlighting” the image with a light source

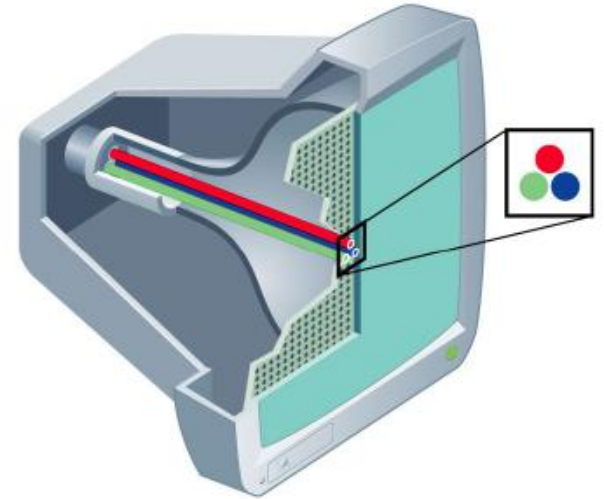
- TV sets are CRTs and many desktop monitors use this technology

- LCD is primarily used for laptops and other portable devices



Video Display Terminology

- Pixel
 - picture element (smallest unit of an image, basically a single dot on the screen)
- Resolution
 - number of pixels in the image
 - Common resolution size is 1024x768
 - Refresh rate
 - how often a CRT's electron gun rescans
 - LCD displays do not use an electron gun, so do not perform



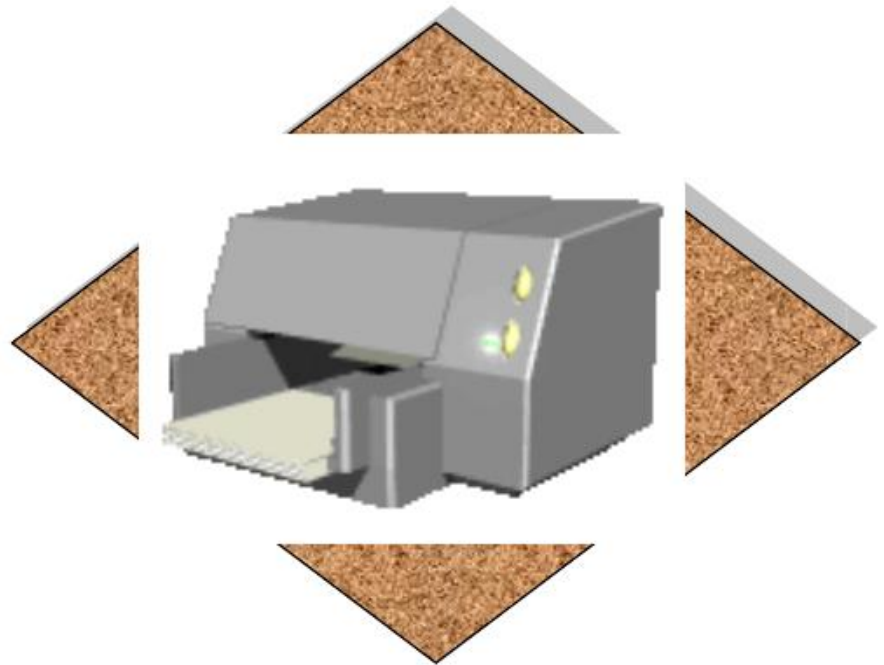
The CRT electron gun “shoots” 3 electrons at the screen representing the amount of red, green and blue for the *pixel*

Printers

- Ink Jet Printer
 - least expensive, color, slower with a higher per page cost than laser printers
- Laser Printer
 - More expensive, faster, lower per page cost than ink jet,

Printers

- A printer is a peripheral device that produces a physical copy or **hard copy** of the computer's output.



Types of Printers

Inkjet



- **Inkjet printer**, also called a bubble-jet, makes characters by inserting dots of ink onto paper
- Letter-quality printouts
- Cost of printer is inexpensive but ink is costly

Laser



- **Laser printer** works like a copier
- Quality determined by dots per inch (dpi) produced
- Color printers available
- Expensive initial costs but cheaper to operate per page

Plotter



- A plotter is a printer that uses a pen that moves over a large revolving sheet of paper.
- It is used in engineering, drafting, map making, and seismology.

Audio Output: Sound Cards and Speakers

- Audio output is the ability of the computer to output sound.
- Two components are needed:
 - **Sound card** – Plays contents of digitized recordings
 - **Speakers** – Attach to sound card

Summary

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THANK YOU